

Bagging vs Boosting

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Underfitting ([Note 91](#)); Ensemble learning ([Note 101](#)); Bias-variance trade-off ([Note 109](#)).
- **Leads to:** Gradient boosting ([Note 120](#)).
- **Compare with:** Bagging ([Note 107](#)); Stacking and blending ([Note 127](#)).

1. Overview

Key point: Bagging and boosting differ on three points: the kind of base model (low bias and high variance, against high bias and low variance), the way they learn (in parallel, against in sequence), and the weight of each base model in the vote (equal, against earned).

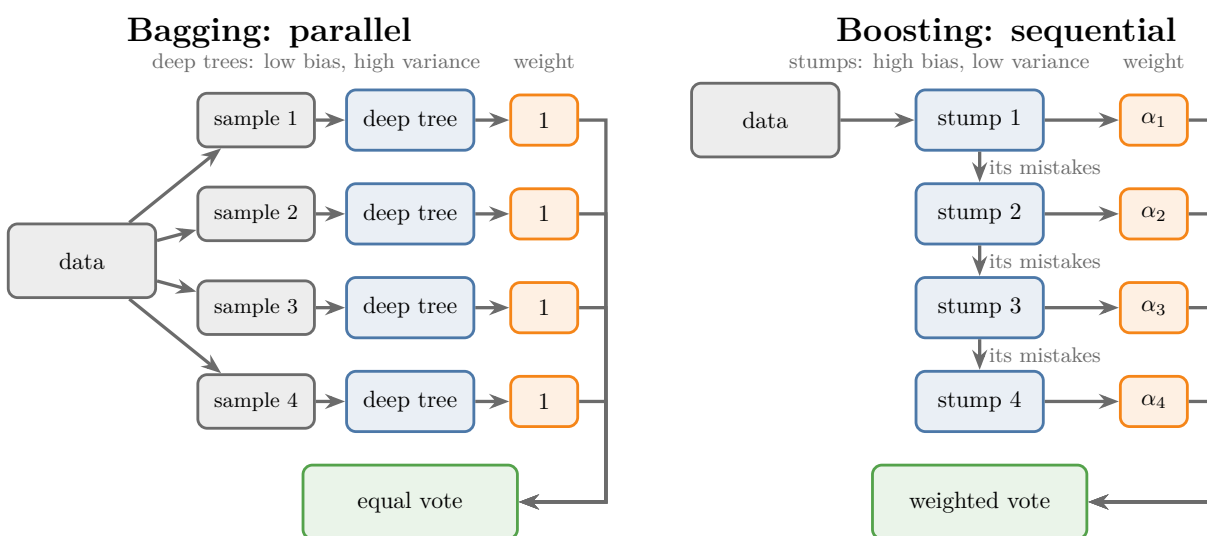


Figure 1: Bagging trains deep trees side by side on random samples and gives each an equal vote; boosting trains stumps one after another, each on the previous one's mistakes, and weights each vote by its alpha

On tables of data, tree ensembles built by bagging (random forests) and by boosting (gradient boosting, XGBoost) are among the strongest models: a benchmark on 45 medium-sized datasets found them ahead of deep networks (Grinsztajn et al. 2022). “What is the difference between bagging and boosting?” is also a common interview question.

We have now seen both: bagging in the [bagging Note](#) and random forests, boosting through AdaBoost in the [AdaBoost intuition Note](#). The present Note puts them side by side on three

points (Figure 1). The Notebook ([notebook.ipynb](#)) runs a small experiment on the first point.

2. Difference 1: the type of base model

Key point: Bagging starts from low-bias, high-variance models and lowers their variance; boosting starts from high-bias, low-variance models and lowers their bias.

We want low bias and low variance, but a single model usually trades one for the other (the [bias-variance Note](#)). Bagging takes **low-bias, high-variance** models, such as fully grown trees or KNN with a small k , and averages away much of their variance (the [bagging Note](#), section 3.2). Boosting attacks the trade-off from the opposite end.

2.1 Boosting: high bias, low variance models

Key point: Boosting uses models that are too simple to fit the data well but are stable, such as shallow trees and decision stumps.

Boosting needs base models with **high bias and low variance**:

- a **shallow decision tree**, whose depth is very small;
- above all the **decision stump**, a tree of depth 1 (the AdaBoost intuition Note, section 2.2).

A stump is not good on the training data, but it hardly changes when the data changes. Adding many stumps in sequence, each fixing the last one's mistakes, lowers the bias step by step while the variance stays fairly low: on noisy data it creeps up only with very many stumps (average test accuracy 0.846 with 50 stumps, 0.837 with 1,500, in the [AdaBoost hyperparameters Note](#), section 3). ESL §15.2 states the same contrast: bagged trees improve only through lower variance, while boosting grows its trees adaptively to remove bias.

2.2 The rule of thumb

Key point: A model that is very good on the training data but unstable calls for bagging; a model that is stable but weak calls for boosting.

If the algorithm is...	it is...	so use
very good on the training data, unstable when the data changes	low bias, high variance	bagging
not so good on the training data, stable when the data changes	high bias, low variance	boosting

The type of base model is the most important of the three differences.

Extra: Swapping the base models shows why. On the noisy circles data of the [AdaBoost hyperparameters Note](#), with 100 base models each and 10-fold cross-validation:

Base model	Alone	Bagging	AdaBoost
stump	0.566	0.738	0.814
fully grown tree	0.746	0.808	0.750

Boosting helps stumps most, bagging helps full trees most. AdaBoost with full trees is no better than one tree: the first tree already classifies every training observation correctly, its error is 0, and boosting stops after that single tree, since no mistakes are left to pass on.

3. Difference 2: parallel against sequential learning

Key point: Bagging trains all its models side by side, independently; boosting trains them one after another, each depending on the one before.

Bagging is parallel (Figure 1, left). Bagging stands for bootstrap aggregation (the bagging Note). From a dataset of, say, 1,000 **observations** (records, one row of the data table each), each base model gets its own random sample of observations. No model needs anything from any other, so all of them can be trained at the same time.

Boosting is sequential (Figure 1, right). The data goes to model 1, which does its job and passes its mistakes on to model 2. Model 2 does its job and passes its own mistakes to model 3, and so on. Model 2 cannot start before model 1 has finished, because it learns from model 1's mistakes.

Extra: Parallel training has a practical side. Bagging can spread its models over every CPU core (`n_jobs=-1` in `BaggingClassifier` and `RandomForestClassifier`); `AdaBoostClassifier` has no `n_jobs`, because its stages must run one at a time.

4. Difference 3: the weight of each base model

Key point: In bagging every base model's vote counts the same, like a democracy; in boosting each model's vote is weighted by how well it did.

When a new query point arrives, every trained base model gives its answer. The two methods combine those answers differently.

Bagging: equal votes. Suppose four models answer 1, 1, 0 and 1. Every vote has the same weight, like a democracy, so the majority wins: three say 1, the output is 1 (Figure 1, left: every weight is 1).

Boosting: weighted votes. Every model carries its own weight, its alpha (the [AdaBoost step-by-step Note](#), section 6). With weights such as 0.8, 1.5 and 5, the model with weight 5 is listened to far more than the one with 0.8. A model that made fewer mistakes during

training earns a larger say.

5. Summary

	Bagging	Boosting
Base models	low bias, high variance (fully grown trees, KNN with small k)	high bias, low variance (stumps, shallow trees)
Mainly reduces Learning	variance parallel: models trained independently, at the same time	bias sequential: each model learns from the previous one's mistakes
Data each model sees	a random sample of the observations	all observations, reweighted toward earlier mistakes
Weight in the vote	equal (democracy)	each model's alpha, earned by its accuracy
Examples	bagging, random forest	AdaBoost, gradient boosting, XGBoost

- Both aim at low bias and low variance, from opposite ends.
- Bagging averages unstable, accurate models; boosting adds up stable, weak ones.
- Bagging trains in parallel and votes equally; boosting trains in sequence and weights each vote.

6. Sources

- Grinsztajn, L., Oyallon, E. and Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *NeurIPS 2022, Datasets and Benchmarks Track*. arxiv.org/abs/2207.08815
- Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning* (ESL), 2nd ed. Springer. §15.2 (bagging reduces variance only; boosting removes bias).

7. Key terms

Term	Meaning
Parallel learning	Training the base models independently, so they can all be trained at once (bagging)
Sequential learning	Training the base models one after another, each depending on the previous ones (boosting)
Shallow decision tree	A decision tree with a small maximum depth; high bias, low variance